

TECHNICAL ASSESMENT REPORT

Automated Property Data Integration & Listing Simulation

Target Company : Bali Home Immo (bali-home-immo.com)
Position : Web Developer
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Date : 2026

1. INTRODUCTION & TECHNICAL APPROACH

This report is submitted as part of the Web Developer recruitment selection at Bali Home Immo. In response to the property listing simulation task, I approached this challenge through a full-stack developer lens, combining automated data aggregation with an interactive frontend demonstration. The deliverables for this assessment include this PDF report documenting the data collection methodology, structured property listings, and backend JSON data mappings, as well as a fully functional, responsive HTML demo page built with vanilla HTML, CSS, and JavaScript. The demo page features image sliders, property cards, and inline JSON code blocks that mirror the Bali Home Immo visual identity. Rather than relying on manual copy pasting, the property data presented here is simulated as the output of an automated parsing pipeline using Python or Node.js, which extracts raw listings from public marketplace channels, normalizes the fields, and outputs database-ready JSON objects ready for direct CMS ingestion.

2. DATABASE SCHEMA & DATA STRUCTURE (JSON READY)

To ensure seamless integration with the bali-home-immo.com backend, every property record follows a consistent schema. This is the same schema rendered in the interactive demo page:

```
{
  "property_id": "BHI-UNIQUE-ID",
  "title": "Full Property Title String",
  "location": {
    "area": "Detailed Area Name",
    "regency": "Badung / Tabanan"
  },
  "price": {
    "idr": 0,
    "usd": 0,
    "type": "Freehold | Leasehold",
    "duration_years": 0,
    "negotiable": true
  },
  "specifications": {
    "bedrooms": 0,
    "bathrooms": 0,
    "land_m2": 0,
    "building_m2": 0,
    "pool": true,
    "furnished": "Full | Semi | Unfurnished"
  },
  "legality": {
    "certificate": "SHM | SHB | Leasehold",
    "additional": "SLF | IMB | In Process"
  },
  "media": {
    "images_cdn": ["url_1.webp", "url_2.webp", "url_3.webp"]
  }
}
```

3. DATA INTEGRATION RESULTS: 5 PREMIUM VILLAS

The following 5 properties are sorted by price (descending) and match exactly with the interactive HTML demo page.

Property 1: Premium Oceanfront Villa - Balian Beach, Tabanan

Field	Value
System Property ID	BHI-TABANAN-001
Property Title	Premium Luxury Oceanfront Villa
Location	Balian Beach, Tabanan, Bali
Legality/Status	Freehold SHM
Selling Price	IDR 14,500,000,000
Bedrooms/Bathrooms	3 BR / 3 BA
Building Size	356 m ²
Key Features	Infinity Pool with panoramic ocean view, DDAP Architect design (2021), 50 m to surf spot, 49 km from Ngurah Rai Airport, Fully Furnished, TV, Wi-Fi, Private Parking
CDN Image Reference	bali1.jpg · bali2.jpg · bali3.jpg

Backend JSON Mapping:

```
{
  "property_id": "BHI-TABANAN-001",
  "price": { "idr": 14500000000, "type": "Freehold" },
  "specs": { "beds": 3, "pool": true, "build_m2": 356 }
}
```

Property 2: Gated Ricefield View Villa - Seminyak, Badung

Field	Value
System Property ID	BHI-SEMINYAK-004
Property Title	Gated Ricefield View Villa
Location	Seminyak, Badung, Bali
Legality/Status	Freehold Hak Milik
Selling Price	IDR 3,900,000,000 (Negotiable)
Bedrooms/Bathrooms	2 BR / 2 BA
Building Size	112 m ² / 100 m ²
Key Features	North-facing, dual-side ricefield views, gated community with 24-hour security, 6-meter access road, Wi-Fi included, Fully Furnished
CDN Image Reference	bali4.jpg · bali5.jpg · bali6.jpg

Backend JSON Mapping:

```
{
  "property_id": "BHI-SEMINYAK-004",
  "price": { "idr": 39000000000, "nego": true },
  "specs": { "beds": 2, "gated": true, "view": "ricefield" }
}
```

Property 3: Mediterranean Elegance Exclusive Villa – Ungasan, South Kuta

Field	Value
System Property ID	BHI-UNGASAN-002
Property Title	Mediterranean Elegance Exclusive Villa
Location	Ungasan, South Kuta, Badung, Bali
Legality/Status	Leasehold 25 Years
Selling Price	IDR 3,350,000,000 (≈ USD 210,000)
Bedrooms/Bathrooms	3 BR / 3 Ensuite BA
Building Size	140 m ² / 120 m ²
Key Features	Private Pool, 100 m to Jimbaran-Uluwatu main road, 10 mins to GWK, Balangan, Dreamland & Bingin Beach, 5500 VA electricity, PDAM water, Fully Furnished
CDN Image Reference	bali7.jpg · bali8.jpg · bali9.jpg

Backend JSON Mapping:

```
{
  "property_id": "BHI-UNGASAN-002",
  "price": { "idr": 33500000000, "usd": 210000 },
  "lease": { "years": 25, "renewable": true }
}
```

Property 4: New Santorini Style Villa – Nusa Dua Elite Cluster

Field	Value
System Property ID	BHI-NUSADUA-005
Property Title	New Santorini Style Villa — Elite Cluster
Location	Nusa Dua, Elite Cluster, Badung, Bali
Legality/Status	Freehold SHM + SLF
Selling Price	IDR 2,300,000,000 (Negotiable)
Bedrooms/Bathrooms	2 BR / 2 BA
Building Size	129 m ² / 90 m ²
Key Features	Mediterranean Santorini design, Private Pool, Elite gated cluster, Rental yield potential IDR 200M/year, SLF Certified, 10 mins to Gunung Payung Beach, 15 mins to Sundays Beach Club, 20 mins to GWK
CDN Image Reference	bali10.jpg · bali11.jpg · bali12.jpg

Backend JSON Mapping:

```
{
  "property_id": "BHI-NUSADUA-005",
  "price": { "idr": 2300000000, "nego": true },
  "rental_yield_idr_annual": 200000000
}
```

Property 5: Brand New Santorini Style Villa – Melasti, Ungasan

Field	Value
System Property ID	BHI-UNGASAN-006
Property Title	Brand New Villa — Santorini Style Melasti
Location	Melasti, Ungasan, Badung, Bali
Legality/Status	Freehold SHM (SLF in process)
Selling Price	IDR 1,950,000,000 (Negotiable)
Bedrooms/Bathrooms	2 BR / 3 BA
Building Size	130 m ² / 100 m ²
Key Features	Brand new Santorini-style architecture, Swimming Pool, Modern Kitchen Set (Free), Garden & Balcony, Carport, 5500W electricity, PDAM water, 5m road access
CDN Image Reference	bali13.jpg · bali14.jpg · bali15.jpg

Backend JSON Mapping:

```
{
  "property_id": "BHI-UNGASAN-006",
  "price": { "idr": 1950000000, "nego": true },
  "specs": { "beds": 2, "baths": 3, "land_m2": 130,
    "build_m2": 100 }
}
```

4. INTERACTIVE DEMO & TECHNICAL HIGHLIGHTS

In addition to this report, I built a fully functional interactive listing page (index.html) that demonstrates vanilla HTML, CSS, and JavaScript implementation with no frameworks, ensuring maximum compatibility and fast load times. Each property card features a custom built image slider with navigation controls and dot indicators, while inline JSON code blocks bridge the gap between visual presentation and database structure. The responsive design adapts seamlessly from desktop to mobile viewports.

5. CONCLUSION

By delivering both a comprehensive data report and an interactive frontend prototype, this assessment demonstrates my capability to automate data collection using scripting pipelines, structure property data into consistent database-ready JSON schemas, build pixel-perfect responsive UI components that match the Bali Home Immo design language, and bridge the gap between raw data aggregation and polished frontend

presentation. I am excited about the opportunity to bring these skills to the Bali Home Immo engineering team, optimizing data workflows and enhancing the user experience on bali-home-immo.com.